

4.7.51

EE25BTECH11054
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Question(4.7.51) Find the equation of the plane passing through $(-1, 3, 2)$ and perpendicular to the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$.

Solution:

Normals of the given planes are

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix}. \quad (0.1)$$

Let the required plane have normal vector $\mathbf{n}$

Since it is perpendicular to both given planes:

$$\mathbf{n}_1^\top \mathbf{n} = 0, \quad \mathbf{n}_2^\top \mathbf{n} = 0. \quad (0.2)$$

That is,

$$\begin{pmatrix} \mathbf{n}_1 & \mathbf{n}_2 \end{pmatrix}^\top \mathbf{n} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.3)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 3 & 1 \end{pmatrix} \mathbf{n} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (0.4)$$

Let

$$\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad (0.5)$$

$$a + 2b + 3c = 0, \quad (0.6)$$

$$3a + 3b + c = 0. \quad (0.7)$$

From these, we get

$$\mathbf{n} = t \begin{pmatrix} 7 \\ -8 \\ 3 \end{pmatrix}, \quad t \in \mathbb{R}, \quad t \neq 0 \quad (0.8)$$

Equation of plain is

$$\mathbf{n}^\top \mathbf{x} = 1 \quad (0.9)$$

since point $\mathbf{p} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ lies on the plain

$$\mathbf{n}^\top \mathbf{p} = 1 \quad (0.10)$$

Substituting,

$$\begin{pmatrix} 7 & -8 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 & -8 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}. \quad (0.11)$$

$$\begin{pmatrix} 7 & -8 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -25 \quad (0.12)$$

$$\frac{-1}{25} \begin{pmatrix} 7 & -8 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 1 \quad (0.13)$$

$$\mathbf{n} = \frac{-1}{25} \begin{pmatrix} 7 \\ -8 \\ 3 \end{pmatrix} \quad (0.14)$$

$$\mathbf{n}^\top \mathbf{x} = 1. \quad (0.15)$$

Intersection of Planes and Required Perpendicular Plane

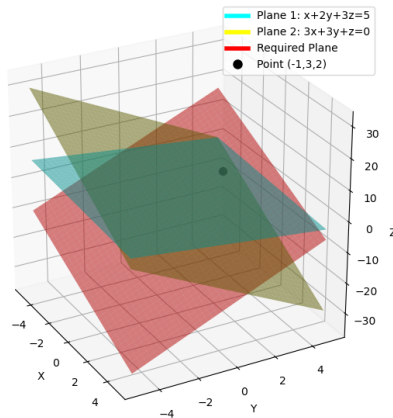


Fig. 0.1